SYMPTOM APPROACH SHEET · 01

Chest Pain

From the complaint → to the diagnosis → to the cause → to the cells → to the treatment

CONTENTS

1	The first 10 minutes	2
2	The map: where chest pain comes from	4
3	History: questions that split the map	6
4	Examination: what each sign means	9
5	Investigations, ECG atlas and scores	11
6	Illness scripts: every cause side by side	17
7.1	Acute Coronary Syndrome	19
7.2	Aortic Dissection	31
7.3	Pulmonary Embolism	34
7.4	Tension Pneumothorax	38
7.5	Cardiac Tamponade	39
7.6	Esophageal Rupture and Perforated Ulcer	40
7.7	Pericarditis and Myocarditis	42
7.8	Stable and Vasospastic Angina	45
7.9	The benign causes	46
★	Credits and sources	48

THE ONE IDEA OF THIS SHEET

Chest pain is a **rule-out** complaint before it is a diagnose complaint. Do not start by asking "what is the most likely cause?". Start by asking **"which of the killers could this be, and how do I exclude each one?"**. Only when the killers are excluded do you look for the common benign cause.

1 The first 10 minutes

EMERGENCY ROOM

About 6% of emergency patients with chest pain have a life-threatening cause, and more than 90% of those are Acute Coronary Syndrome (ACS). Patients with a dangerous cause can look deceptively well, so the first minutes are protocol, not intuition.

Acute Coronary Syndrome (ACS)
Pressure-like pain, radiation to arm/jaw, sweating. electrocardiogram (ECG) and troponin.

Aortic Dissection
Sudden, maximal at onset, tearing, to the back. Pulse or blood pressure difference between arms.

Pulmonary Embolism (PE)
Pleuritic pain + breathlessness + tachycardia + low oxygen saturation.

Tension Pneumothorax
Absent breath sounds on one side, hyperresonance, hypotension, raised jugular venous pressure.

Cardiac Tamponade
Hypotension, raised jugular venous pressure, muffled heart sounds (Beck triad).

Esophageal Rupture / Perforated Ulcer
Pain after vomiting, subcutaneous emphysema; or rigid abdomen.

Also dangerous, less common: myocarditis, stress (takotsubo) cardiomyopathy. Both look like acute coronary syndrome (ACS) on electrocardiogram (ECG) and troponin.

1

Look at the patient and the vital signs

Airway, Breathing, Circulation (ABC). Heart rate, blood pressure **in both arms**, respiratory rate, oxygen saturation, temperature, glucose.

Unstable (shock, hypoxia, altered consciousness): resuscitate, call senior, bedside ultrasound now.

Stable: continue the ladder, but keep the patient on a monitor with a defibrillator nearby.

2

12-lead electrocardiogram (ECG) within 10 minutes of arrival

Repeat it if the pain changes, and compare with old ECGs.

ST elevation or STEMI-equivalent → ST-elevation myocardial infarction (STEMI) pathway: reperfusion now (section 7.1).

No ST elevation → it is NOT excluded. Continue with troponin and the other killers.

3

Start general measures while you think

intravenous (IV) access + bloods (high-sensitivity troponin, full blood count, kidney function, electrolytes, coagulation, lipase, D-dimer only if indicated). Oxygen **only** if saturation <90%. Pain relief.

4

Focused history and examination aimed at the six killers

Sections 3 and 4. Every question and every sign below is chosen because it moves one killer up or down.

5

Chest X-ray (CXR) for most patients

Can confirm pneumothorax, pneumonia, free air under the diaphragm, widened mediastinum. A normal CXR does **not** exclude acute coronary syndrome (ACS), pulmonary embolism (PE) or dissection.

6

Targeted tests for the killer you still suspect

Serial troponin (acute coronary syndrome (ACS)), Wells → D-dimer or computed tomography (CT) pulmonary angiography (pulmonary embolism (PE)), Aortic Dissection Detection Risk Score (ADD-RS) → D-dimer or CT aortogram (dissection), echo (tamponade). Section 5.

ASSUME A KILLER UNTIL PROVEN OTHERWISE IF

- Abnormal vital signs, obvious distress, or signs of poor perfusion (sweaty, cold, clammy).
- Abrupt pain that is sharp, tearing or ripping, in the chest or abdomen.
- Missing pulse, or systolic blood pressure difference >20 mmHg between the two arms.
- ST elevation on the electrocardiogram (ECG).

2 The map

WHERE CHEST PAIN COMES FROM

Think in **anatomy**: every structure inside or on the chest can hurt. Walk the map from the inside out: heart → aorta → lungs and pleura → esophagus and upper abdomen → chest wall → mind.

CHEST PAIN

HEART

Ischemic:
- **Acute Coronary Syndrome**

Stable angina
Vasospastic angina
Non-ischemic:
- **Cardiac tamponade**
- **Myocarditis**
- **Takotsubo cardiomyopathy**

Pericarditis
Aortic stenosis, HCM

AORTA

- **Aortic dissection**

Expanding / leaking thoracic aneurysm

LUNG & PLEURA

- **Pulmonary embolism**
- **Tension pneumothorax**

Simple pneumothorax
Pneumonia, pleurisy
Pulmonary hypertension

GUT

- **Esophageal rupture**
- **Perforated peptic ulcer**

Reflux / esophagitis
Esophageal spasm
Biliary colic, cholecystitis
Pancreatitis

CHEST WALL & NERVES

Costochondritis
Rib fracture, trauma
Muscle strain
Herpes zoster
Thoracic nerve root pain

MIND

Panic attack
Anxiety, somatic pain
(only after the killers are excluded)

Red = can kill within hours. Rule these out first.

Why the pain feels the way it does (the physiology behind the questions)

VISCERAL PAIN → DULL, POORLY LOCALIZED

Heart, aorta and esophagus have few sensory nerve fibres that enter the spinal cord over several segments (T1–T5). The brain cannot locate them, so the pain is **diffuse, pressure-like, central**, and the patient often points with a fist (Levine sign).

These fibres **share spinal neurons** with skin nerves from the chest, inner arm (T1–T2), neck and jaw. The brain misreads the signal as coming from the skin → **referred pain** to the left arm, jaw, shoulders.

SOMATIC PAIN → SHARP, LOCALIZED

The **parietal pleura**, parietal pericardium and chest wall are supplied by intercostal nerves (somatic). Their pain is **sharp, well localized, and worse when the structure moves**: breathing (pleuritic), changing position, pressing.

So **pleuritic pain = parietal pleura or pericardium is involved**: pulmonary embolism (PE) with infarction, pneumonia, pneumothorax, pericarditis. The diaphragmatic pericardium is supplied by the phrenic nerve (C3–C5) → pericarditis pain to the **trapezius ridge**.

3 History

QUESTIONS THAT SPLIT THE MAP

Every question has a reason. The right column tells you which branch of the map the answer pushes you toward.

Characterize the pain (Site, Onset, Character, Radiation, Associations, Time, Exacerbating/relieving factors, Severity (SOCRATES))

Ask	Answer	Points to
Onset How did it start?	Sudden, maximal within seconds	**Aortic dissection, pulmonary embolism (PE), pneumothorax**
	Builds over minutes	acute coronary syndrome (ACS)
	After forceful vomiting	**Esophageal rupture (Boerhaave)**
Character	Pressure, heaviness, tightness, squeezing	ACS, stable angina
	Tearing, ripping	**Aortic dissection**
	Sharp, stabbing, pleuritic	PE, pneumothorax, pneumonia, pericarditis
	Burning, behind the sternum	Reflux (gastroesophageal reflux disease (GERD)), esophagitis
	Burning in a band on one side, then rash	Herpes zoster

Ask	Answer	Points to
Radiation	Left or **both** arms, jaw, neck	acute coronary syndrome (ACS)
	To the back, between the shoulder blades, moving down	**Aortic dissection**
	To the trapezius ridge (top of shoulder)	Pericarditis
	To the back from the epigastrium	Pancreatitis, penetrating ulcer
Exacerbating / relieving	Exertion brings it on, rest relieves within minutes	Stable angina
	At rest, early morning, young smoker or cocaine	Vasospastic angina
	Worse lying flat, better sitting forward	Pericarditis
	Worse with breathing or coughing	Pleural cause (pulmonary embolism (PE), pneumonia, pneumothorax), pericarditis
	After meals, lying down, bending; better with antacid	gastroesophageal reflux disease (GERD)
Timing	>20 minutes at rest / <10 minutes with exertion / seconds only	ACS / stable angina / almost never cardiac
Severity	"Worst ever", from the first second	**Aortic dissection**

Associated symptoms

Symptom	Points to
Sweating, nausea, vomiting	acute coronary syndrome (ACS) (autonomic response)
Breathlessness	pulmonary embolism (PE), pneumothorax, heart failure from ACS, tamponade
Coughing blood (hemoptysis)	PE with infarction, pneumonia, malignancy
Fainting (syncope)	**Massive PE, dissection with tamponade, arrhythmia from ACS, aortic stenosis**
Stroke symptoms, leg weakness, abdominal pain with the chest pain	**Aortic dissection blocking a branch artery**
Fever, viral illness 1–2 weeks before	Pericarditis, myocarditis, pneumonia
Calf pain or swelling	Deep vein thrombosis → PE
Palpitations, tingling of fingers and lips, fear of dying	Panic attack (only after killers excluded)

Risk factors: whose pain is it?

Disease	Ask about
acute coronary syndrome (ACS)	Age, male sex, smoking, diabetes, hypertension, high cholesterol, family history of early coronary disease, known coronary disease, chronic kidney disease, cocaine
Aortic dissection	Hypertension (most common), Marfan or other connective tissue disease, bicuspid aortic valve, known aneurysm, recent aortic procedure, cocaine, pregnancy
pulmonary embolism (PE)	Recent surgery or immobilization, cancer, previous clot, estrogen (pill, pregnancy, postpartum), long travel, known clotting disorder
Pneumothorax	Tall thin young man, smoker, chronic obstructive pulmonary disease (COPD), recent chest procedure or trauma
Pericarditis	Recent viral illness, recent heart attack or cardiac surgery, kidney failure (uremia), autoimmune disease, tuberculosis exposure, cancer

EVIDENCE: HOW MUCH EACH FEATURE CHANGES THE PROBABILITY OF ACUTE CORONARY SYNDROME (ACS)

Likelihood ratio (LR) >1 makes ACS more likely, <1 less likely. Values from the meta-analysis by Swap and Nagurney (JAMA 2005).

↑ More likely

- Radiation to right arm or shoulder: likelihood ratio (LR) 4.7
- Radiation to both arms: LR 4.1
- Brought on by exertion: LR 2.4
- Radiation to left arm: LR 2.3
- Sweating: LR 2.0
- Nausea or vomiting: LR 1.9

↓ Less likely

- Pleuritic: likelihood ratio (LR) 0.2
- Positional: LR 0.3
- Sharp, stabbing: LR 0.3
- Reproduced by pressing on the chest: LR 0.3

Even likelihood ratio (LR) 0.2–0.3 does not exclude acute coronary syndrome (ACS). A patient can have a tender chest wall AND a heart attack.

ATYPICAL PRESENTERS: THE ONES WHO GET MISSED

Women, elderly, people with diabetes (autonomic neuropathy damages the pain fibres), and patients with chronic kidney disease often have **no classic pain**: only breathlessness, fatigue, epigastric discomfort, nausea or confusion. Do the electrocardiogram (ECG) anyway.

4 Examination

WHAT EACH SIGN MEANS

The examination is often **normal in acute coronary syndrome (ACS)**. Its main job is to catch the other killers and to find complications.

Sign	Meaning	Why (mechanism)
Blood pressure difference >20 mmHg between arms, or absent pulse	**Aortic dissection**	False lumen compresses the subclavian or carotid origin
New early diastolic murmur (left sternal edge)	**Type A dissection → aortic regurgitation**	Dissection reaches the aortic root and detaches the valve support
Absent breath sounds + hyperresonance + trachea pushed away + hypotension	**Tension pneumothorax**	Air under pressure collapses the lung and pushes the mediastinum
Raised jugular venous pressure (JVP) + hypotension + muffled heart sounds	**Tamponade (Beck triad)**	Fluid under pressure stops the right heart filling
Pulsus paradoxus (systolic drop >10 mmHg on inspiration)	Tamponade (also severe asthma)	Inspiration fills the right ventricle, which bulges the septum into the left ventricle
Raised JVP + clear lungs + hypotension	Massive pulmonary embolism (PE), right ventricular infarction, tamponade, tension pneumothorax	Obstruction or failure on the right side
Bibasal crackles, third heart sound	Left ventricular failure (large acute coronary syndrome (ACS))	Back pressure into pulmonary veins → edema
New pansystolic murmur days after myocardial infarction (MI)	**Papillary muscle rupture (mitral regurgitation) or ventricular septal rupture**	Necrotic muscle tears at day 3–14

Sign	Meaning	Why (mechanism)
Pericardial friction rub (scratchy, changes with position)	Pericarditis	Inflamed fibrin-covered layers rub against each other
Unilateral swollen, tender calf	Deep vein thrombosis → pulmonary embolism (PE)	Source of the embolus
Tachycardia + tachypnea + low saturation, normal chest	PE	Ventilated but not perfused lung (dead space)
Subcutaneous emphysema (crackling skin), crunch with heartbeat (Hamman sign)	**Esophageal rupture, pneumothorax**	Air tracking into the mediastinum and neck
Rigid, silent abdomen	**Perforated peptic ulcer**	Chemical peritonitis from gastric acid
Focal tenderness exactly reproducing the pain	Costochondritis, rib fracture	Somatic source in the chest wall
Vesicles in one dermatome (may appear 2–3 days after the pain)	Herpes zoster	Reactivated varicella in one sensory ganglion
Focal neurological deficit with chest pain	**Aortic dissection**	Carotid involvement

BEDSIDE ULTRASOUND (POINT-OF-CARE ULTRASOUND (POCUS)) IN THE UNSTABLE PATIENT

One probe answers four killers: **pericardial effusion** (tamponade), **dilated right ventricle** (massive pulmonary embolism (PE)), **no lung sliding** (pneumothorax), **regional wall not moving** (acute coronary syndrome (ACS)), and a **dilated aortic root or flap** (dissection).

5 Investigations

ECG ATLAS · TROPONIN · SCORES

Which test answers which question

Test	Answers	Remember
electrocardiogram (ECG) (within 10 min, repeat)	ST-elevation myocardial infarction (STEMI), ischemia, pericarditis, arrhythmia, pulmonary embolism (PE) strain pattern	A normal first ECG does not exclude acute coronary syndrome (ACS)
High-sensitivity troponin at 0 h and 1–3 h	Myocardial injury: ACS, also myocarditis, PE, dissection, sepsis, kidney failure	Look at the **change** between samples, not only the value
Chest X-ray	Pneumothorax, pneumonia, effusion, widened mediastinum, free air	Normal in most ACS, PE and many dissections
D-dimer	Excludes PE (low/intermediate probability) or dissection (Aortic Dissection Detection Risk Score (ADD-RS) 0–1)	High sensitivity, low specificity. Use age-adjusted cut-off after 50: age × 10 µg/L
computed tomography (CT) pulmonary angiography	PE	Needs kidney function and contrast check
CT aortography	Aortic dissection (stable patient)	Unstable → transesophageal echo at bedside
Echocardiography	Effusion and tamponade, wall motion, right ventricular strain, valves	Bedside version first if unstable
Coronary angiography	Gold standard for coronary occlusion; also treats it (percutaneous coronary intervention (PCI))	Timing depends on STEMI vs non-ST-elevation acute coronary syndrome (NSTE-ACS) risk
Water-soluble contrast swallow / CT with oral contrast	Esophageal rupture	Not barium first (mediastinitis)
Lipase, liver tests	Pancreatitis, biliary disease	Upper abdomen can mimic chest pain

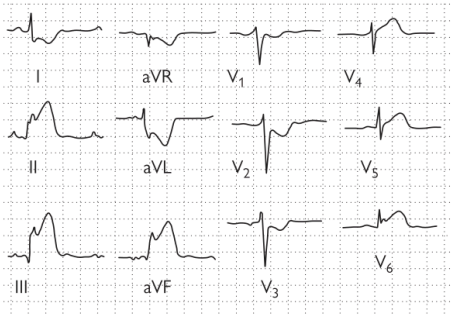
electrocardiogram (ECG) atlas

ST-ELEVATION MYOCARDIAL INFARCTION (STEMI) CRITERIA (FOURTH UNIVERSAL DEFINITION OF MI)

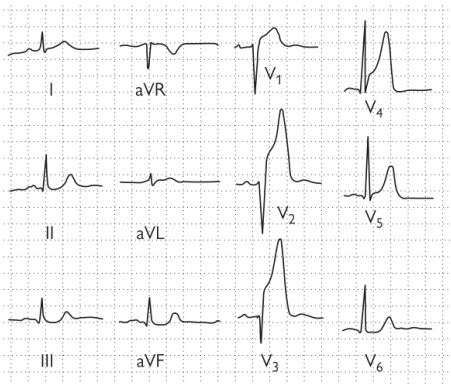
New ST elevation at the J point in **two contiguous leads**: ≥1 mm in all leads except V2–V3, where it must be ≥2 mm in men ≥40 years, ≥2.5 mm in men <40 years, ≥1.5 mm in women. **STEMI-equivalents** (treat as STEMI): new or presumed new left bundle branch block with ischemic symptoms, posterior MI (ST depression V1–V3 with ST elevation V7–V9), ST elevation in aVR with widespread ST depression (left main or multivessel disease), de Winter T waves.



Evolution of the electrocardiogram (ECG) in ST-elevation myocardial infarction (STEMI). Hours: hyperacute T, then ST elevation. Days: Q wave forms, ST falls, T inverts. Months: Q wave remains as a scar marker. *Oxford Handbook of Emergency Medicine, Fig. 3.5.*



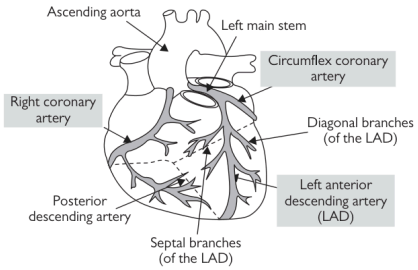
Inferolateral ST-elevation myocardial infarction (STEMI). ST elevation in II, III, aVF (inferior) and V5–V6 (lateral), with reciprocal ST depression in I, aVL and V2–V3. Usually right coronary artery. *Oxford Handbook of Emergency Medicine (OHEM), Fig. 3.6.*



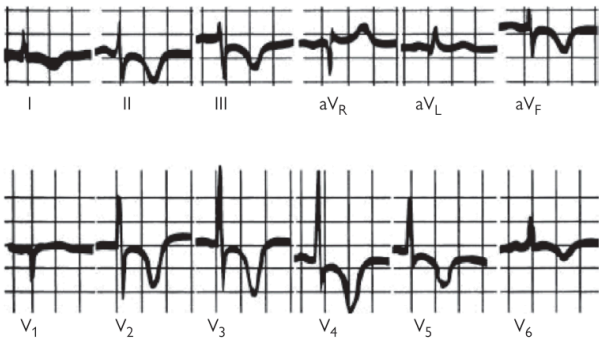
Anteroseptal ST-elevation myocardial infarction (STEMI). ST elevation V1–V4, minimal reciprocal change in III and aVF. Left anterior descending artery. *Oxford Handbook of Emergency Medicine (OHEM), Fig. 3.7.*

Leads → wall → artery

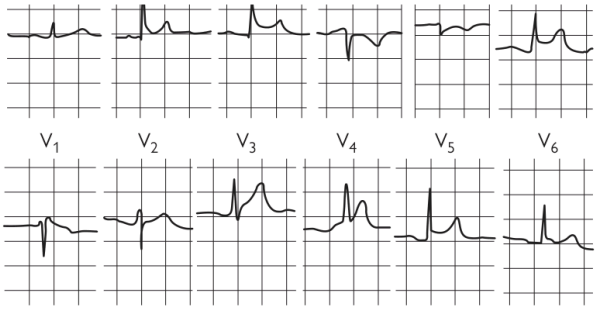
Leads with ST elevation	Wall	Usual artery
V1–V2	Septal (anteroseptal)	Left anterior descending (left anterior descending artery (LAD))
V3–V4	Anterior / apical	LAD
V5–V6	Anterolateral	LAD or left circumflex (left circumflex artery (LCX))
I, aVL	High lateral	LCX (or diagonal branch of LAD)
II, III, aVF	Inferior	Right coronary artery (RCA) in ~80%, LCX in the rest
V1 + right-sided V4R	Right ventricle	Proximal RCA
ST *depression* V1–V3, ST elevation V7–V9	Posterior	Posterior descending artery (RCA or LCX)



Blood supply to the heart. *Oxford Handbook of Emergency Medicine (OHEM), Fig. 3.8.*



Subendocardial ischemia / non-ST-elevation myocardial infarction (NSTEMI). Widespread ST depression and T-wave inversion, no ST elevation. The subendocardium is the last place the blood reaches, so it suffers first. *Oxford Handbook of Emergency Medicine (OHEM), Fig. 3.9.*



Acute pericarditis. Concave ("saddle") ST elevation in almost every lead, PR depression (PR elevation in aVR), no reciprocal changes and no Q waves. *Oxford Handbook of Emergency Medicine (OHEM), Fig. 3.10.*

electrocardiogram (ECG) clue	ST-elevation myocardial infarction (STEMI)	Pericarditis
ST shape	Convex ("tombstone") or straight	Concave ("saddle")
Distribution	One coronary territory	Widespread, many territories
Reciprocal ST depression	Yes	No (except aVR and V1)
PR segment	Normal	Depressed (elevated in aVR)
Q waves	Develop	Never

PULMONARY EMBOLISM (PE) ON THE ELECTROCARDIOGRAM (ECG)

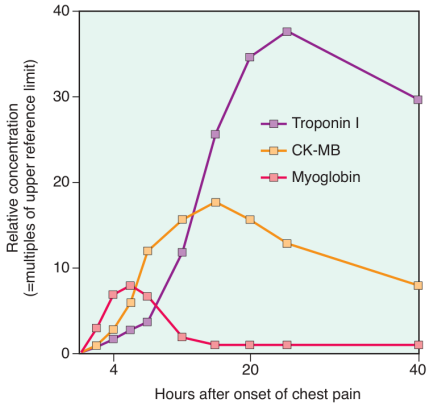
Most common: **sinus tachycardia**. Right heart strain: right bundle branch block, right axis deviation, T inversion V1–V4, S wave in I + Q wave and inverted T in III (S1Q3T3, specific but uncommon). The ECG is used to look for other causes, not to diagnose PE.

TAMPONADE ON THE ELECTROCARDIOGRAM (ECG)

Low-voltage QRS and **electrical alternans** (QRS height alternates beat to beat as the heart swings in the fluid). Sinus tachycardia.

Troponin: reading the curve

- Troponin I and T are structural proteins of the cardiac sarcomere. They leak only when myocytes are injured.
- **High-sensitivity assays** detect a rise within **1–3 hours**, peak at 24–48 h, and stay raised for **7–10 days** (troponin T up to 14 days).
- **Myocardial infarction** = rise and/or fall of troponin with at least one value above the 99th percentile **plus** evidence of ischemia (symptoms, electrocardiogram (ECG) changes, new Q waves, imaging, or thrombus at angiography).
- Rapid protocols (European Society of Cardiology, ESC): 0 h / 1 h or 0 h / 2 h samples. Very low at 0 h with pain >3 h, or low with no change at 1 h → rule out. High at 0 h or significant rise → rule in. In between → observe and repeat.
- **creatine kinase-MB (CK-MB)** falls to normal within 48–72 h, so it helps detect **re-infarction** while troponin is still raised.



Troponin I, creatine kinase-MB (CK-MB) and myoglobin after myocardial infarction (MI).
Robbins Basic Pathology, Fig. 11.13.

Scores that turn suspicion into a decision

History, electrocardiogram (ECG), Age, Risk factors, Troponin (HEART) score: risk of a major cardiac event in 6 weeks (chest pain, no ST-elevation myocardial infarction (STEMI))	
History	Highly suspicious 2 · moderately 1 · slightly 0
ECG	Significant ST deviation 2 · non-specific repolarization changes 1 · normal 0
Age	≥65 years 2 · 45–64 1 · <45 0
Risk factors	≥3 risk factors or known atherosclerotic disease 2 · 1–2 risk factors 1 · none 0
Troponin	>3× normal limit 2 · 1–3× 1 · normal 0
Total	**0–3 low** (about 2% events: discharge with follow-up if serial troponin negative) · **4–6 intermediate** (admit, further testing) · **7–10 high** (early invasive)

Wells score for pulmonary embolism (PE)	
Signs of deep vein thrombosis	3
PE is the most likely diagnosis	3
Heart rate >100	1.5
Immobilization ≥3 days or surgery in last 4 weeks	1.5
Previous DVT or PE	1.5
Hemoptysis	1
Active cancer	1
≤4: PE unlikely → D-dimer. **>4: PE likely** → computed tomography (CT) pulmonary angiography directly.	

Pulmonary Embolism Rule-out Criteria (PERC) rule: use only if clinical probability is already low
Age <50 · heart rate <100 · saturation ≥95% · no hemoptysis · no estrogen use · no previous deep vein thrombosis (DVT)/PE · no unilateral leg swelling · no surgery or trauma needing admission in 4 weeks
All 8 negative → no test needed. Any positive → D-dimer.

ADD-RS: Aortic Dissection Detection Risk Score (1 point per category)
Condition: Marfan or connective tissue disease, family history, known aortic valve disease or aneurysm, recent aortic manipulation
Pain: abrupt, severe, ripping or tearing
Exam: pulse deficit or blood pressure difference, neuro deficit with pain, new aortic regurgitation murmur with pain, hypotension or shock
0–1 + negative D-dimer → dissection unlikely. ≥2 → computed tomography (CT) aortography directly.

6 Illness scripts

EVERY CAUSE SIDE BY SIDE

An illness script is the "typical story" of a disease. Compare the patient in front of you with each script.

The killers

Disease	Typical patient	Pain	Key clue	Confirm with
Acute Coronary Syndrome (ACS)	Older, risk factors	Pressure, central, >20 min, to arm/jaw	Sweating, electrocardiogram (ECG) changes	ECG, serial troponin, angiography
Aortic dissection	Hypertensive man 60s, or young with Marfan	Sudden, maximal at onset, tearing, to the back	Pulse / blood pressure difference, new diastolic murmur	computed tomography (CT) aortography (or transesophageal echo)
Pulmonary embolism (PE)	After surgery, immobility, cancer, estrogen	Pleuritic, sudden	Breathless, tachycardic, low saturation	Wells → D-dimer → CT pulmonary angiography
Tension pneumothorax	Trauma, ventilated patient, tall thin smoker	Sudden, one-sided, pleuritic	Absent breath sounds, hypotension, raised jugular venous pressure (JVP)	**Clinical**: decompress first
Cardiac tamponade	Cancer, kidney failure, after myocardial infarction (MI) or cardiac surgery, trauma	Dull, or that of the cause	Beck triad, pulsus paradoxus	Echo
Esophageal rupture	After heavy vomiting (alcohol)	Severe retrosternal, after vomiting	Subcutaneous emphysema, sick, septic	CT with oral contrast
Perforated peptic ulcer	non-steroidal anti-inflammatory drug (NSAID) use, ulcer history	Sudden epigastric, spreads	Rigid abdomen	Erect chest X-ray (CXR) (free air), CT
Myocarditis	Young, recent viral illness	Variable, may be pericarditic	Heart failure, arrhythmia, raised troponin	Echo, cardiac magnetic resonance imaging (MRI)
Takotsubo cardiomyopathy	Postmenopausal woman after emotional or physical stress	Like ACS	ST elevation, normal coronaries	Angiography + apical ballooning on imaging

The common and benign (diagnosis of exclusion)

Disease	Typical patient	Pain	Key clue	Confirm with
Stable angina	Risk factors	Pressure with exertion, <10 min	Relieved by rest or nitrate	Stress test, computed tomography (CT) coronary angiography
Vasospastic angina	Young smoker, cocaine	At rest, early morning	Transient ST elevation	Clinical, provocation test
Pericarditis	Young, after viral illness	Sharp, pleuritic, better sitting forward	Friction rub, diffuse ST elevation	electrocardiogram (ECG), echo, inflammatory markers
Pneumonia	Any age	Pleuritic	Fever, cough, crackles	chest X-ray (CXR)
Reflux (gastroesophageal reflux disease (GERD))	Obese, after meals	Burning, retrosternal	Sour taste, relieved by antacids	Trial of proton pump inhibitor; endoscopy if alarm features
Esophageal spasm	Any	Squeezing, mimics angina	Dysphagia, cold drinks trigger, relieved by nitrates too	Manometry
Biliary colic, pancreatitis	Fatty meal / alcohol, gallstones	Epigastric, right upper quadrant, to back	Abdominal tenderness	Ultrasound, lipase
Costochondritis	Any	Sharp, focal, with movement	Reproduced by pressing exactly that spot	Clinical
Herpes zoster	Elderly, immunosuppressed	Burning band, one side	Rash in one dermatome (may come later)	Clinical
Panic attack	Young, anxious	Tight, with palpitations	Tingling, hyperventilation, fear	**Only** after the killers are excluded

7.1 Acute Coronary Syndrome (ACS) KILLER

Acute Coronary Syndrome (ACS) is one disease process with three names, depending on how complete the occlusion is and whether the myocardium dies:

Type	What happened	electrocardiogram (ECG) / troponin
Unstable angina	Partial occlusion, ischemia without necrosis	ST depression or T inversion or normal · troponin normal
NSTEMI (non-ST-elevation myocardial infarction)	Partial or transient occlusion → subendocardial necrosis	ST depression or T inversion or normal · **troponin raised**
STEMI (ST-elevation myocardial infarction)	Complete occlusion → transmural necrosis	**ST elevation** · troponin raised

Unstable angina = new angina, angina getting worse (crescendo), or angina at rest. Type 1 myocardial infarction (MI) = plaque rupture (this section). Type 2 MI = supply/demand mismatch without rupture (anemia, tachyarrhythmia, sepsis, hypotension).

RETROGRADE CHAIN: FROM THE PAIN BACK TO THE CELLS

SYMPTOM **Crushing central chest pain radiating to the left arm, sweating, nausea**

▼ WHY? ▼

PHYSIOLOGY Ischemic myocytes switch to anaerobic glycolysis → lactate, H^+, K^+, **adenosine** and bradykinin build up → stimulate cardiac sensory fibres (T1–T5) → referred to chest wall and inner arm. Sympathetic surge → sweating, tachycardia; vagal reflex → nausea

▼ WHY? ▼

HEMODYNAMICS Coronary blood flow stops → oxygen supply < demand. Contraction fails within **60 seconds**; injury becomes irreversible after **20–40 minutes**

▼ WHY? ▼

OCCLUSION Thrombus on top of a ruptured plaque blocks the coronary artery

▼ WHY? ▼

THROMBOSIS Rupture exposes collagen and **tissue factor** → platelets **adhere** (von Willebrand factor → glycoprotein Ib (GP Ib)), **activate** (release adenosine diphosphate (ADP) and thromboxane A2) and **aggregate** (fibrinogen bridges between glycoprotein IIb/IIIa (GP IIb/IIIa) receptors) + coagulation cascade → thrombin → fibrin mesh

▼ WHY? ▼

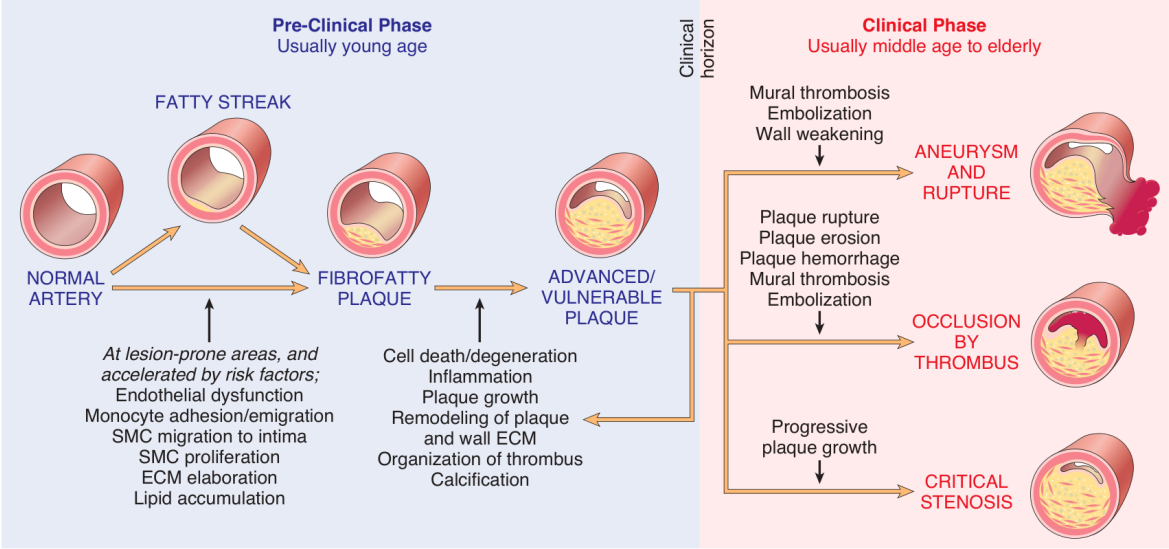
PLAQUE **Vulnerable plaque**: large lipid core, thin fibrous cap, many macrophages. Macrophage enzymes (matrix metalloproteinases) digest the cap → it ruptures under stress (often morning sympathetic surge)

▼ WHY? ▼

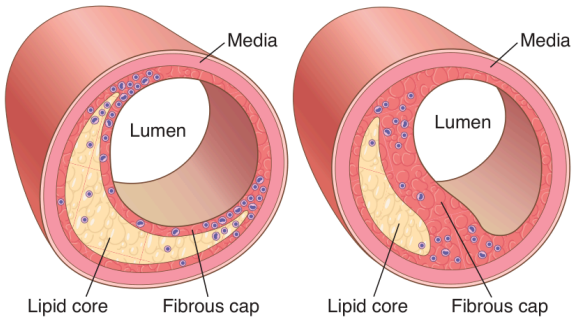
CAUSE **Atherosclerosis: endothelial injury (smoking, hypertension, diabetes, high low-density lipoprotein (LDL)) → LDL enters intima and is oxidized → monocytes enter and become foam cells → fatty streak → smooth muscle cells migrate and make collagen → fibrous cap over a necrotic lipid core**

THE LINK THAT MAKES THE TREATMENT LOGICAL

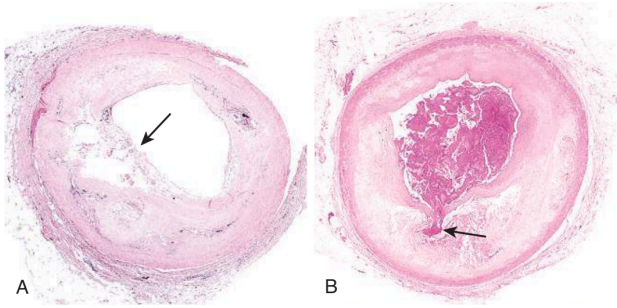
Each drug blocks one step of the thrombosis box above: **aspirin** → thromboxane A2 · **P2Y12 inhibitors** → adenosine diphosphate (ADP) · **heparin** → thrombin and factor Xa · **fibrinolytic** → dissolves fibrin · **percutaneous coronary intervention (PCI)** → mechanically opens the artery · **statin** → stabilizes the plaque · **nitrate and beta-blocker** → reduce the oxygen demand of the starving muscle.



Natural history of atherosclerosis: from normal artery through fatty streak and fibrofatty plaque to the clinical events (rupture, thrombosis, aneurysm, stenosis). *Robbins Basic Pathology, Fig. 10.15.*



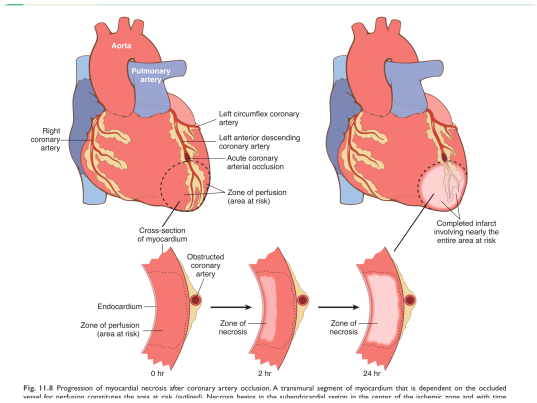
Vulnerable vs stable plaque. Vulnerable: big lipid core, thin cap, inflammation → ruptures → acute coronary syndrome (ACS). Stable: thick cap, small core → narrows slowly → stable angina. *Robbins, Fig. 10.16.*



Plaque rupture. (A) Rupture without thrombus. (B) Rupture (arrow) with an occlusive thrombus that caused a fatal myocardial infarction (MI). *Robbins, Fig. 10.14.*

Where the muscle dies first: the wavefront

- The **subendocardium** is the last layer reached by blood (coronaries run on the outside) and is squeezed hardest in systole → it dies first.
- Necrosis spreads from inside to outside: subendocardial at ~30 min, most of the wall by **3–6 hours**.
- This is why **"time is muscle"**: reperfusion in the first hours saves the outer layers.
- Subendocardial only → ST depression (non-ST-elevation myocardial infarction (NSTEMI)). Full thickness → ST elevation (ST-elevation myocardial infarction (STEMI)).



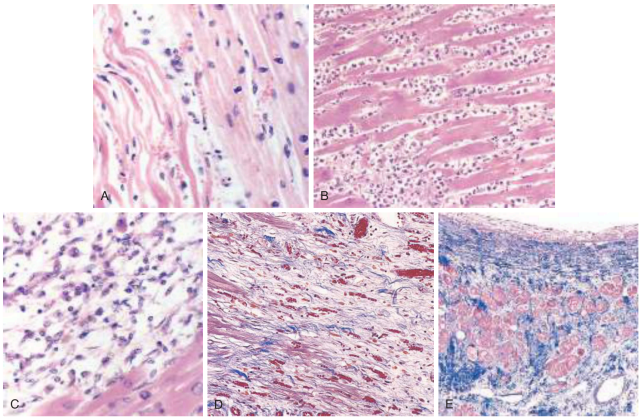
Progression of necrosis after coronary occlusion (0 h → 2 h → 24 h). *Robbins, Fig. 11.8.*

The microscopic clock: what the pathologist sees, and what can go wrong at that time

Time	Gross	Microscopy	Complication at this time
0–4 h	Nothing	Nothing; wavy fibres at the border	**Ventricular fibrillation (most deaths), arrhythmias**
4–24 h	Dark mottling	**Coagulative necrosis** begins: hypereosinophilic cytoplasm, nuclei shrink (pyknosis), contraction bands, edema	**Arrhythmia, heart failure, cardiogenic shock**
1–3 days	Yellow-tan center	Loss of nuclei and striations, **neutrophils** flood in	Early fibrinous pericarditis
3–7 days	Soft yellow, red border	Neutrophils dying, **macrophages** start eating dead fibres	**Wall is softest → rupture: free wall (tamponade), papillary muscle (acute mitral regurgitation), septum (ventricular septal defect)**
7–10 days	Maximally soft	Macrophages at peak, early granulation tissue at margins	
10–14 days	Red-grey borders	**Granulation tissue**: new vessels and collagen	
2–8 weeks	Grey-white scar forming	Collagen increases, cells decrease	Dressler syndrome (autoimmune pericarditis)
>2 months	Firm white scar	Dense collagen **scar**	Ventricular aneurysm, mural thrombus → embolic stroke, heart failure



Acute myocardial infarction (MI), gross. The infarct (arrow) does not take up the triphenyltetrazolium chloride (TTC) stain because its enzymes leaked out. Arrowhead: an old white scar. *Robbins, Fig. 11.10.*



Microscopic clock. (A) Day 1: coagulative necrosis, wavy fibres. (B) Day 2–3: neutrophils. (C) Day 7–10: macrophages clearing dead cells. (D) Granulation tissue. (E) Healed collagen scar (blue, trichrome). *Robbins, Fig. 11.11.*

Management of ST-elevation myocardial infarction (STEMI) (European Society of Cardiology (ESC) 2023)

1

Diagnose: electrocardiogram (ECG) within 10 minutes

→ activate the cath lab.

2

Reperfusion: this is the treatment, everything else supports it

Primary PCI (percutaneous coronary intervention) if it can be done within **120 minutes** of diagnosis (symptoms <12 h). If PCI is not possible within 120 min → **fibrinolysis within 10 minutes** of diagnosis (no contraindication), then transfer to a PCI centre for angiography within 2–24 h (immediately if failed lysis).

3

Antithrombotics

Aspirin 150–300 mg chewed + a P2Y12 inhibitor (prasugrel or ticagrelor preferred; clopidogrel with fibrinolysis or high bleeding risk) + anticoagulant during percutaneous coronary intervention (PCI) (unfractionated heparin).

4

Supportive

Oxygen **only if saturation <90%**. Sublingual or intravenous (IV) nitrate for pain (not if hypotensive, right ventricular infarct, or recent phosphodiesterase-5 (PDE-5) inhibitor). IV opioid for severe pain (it slows absorption of oral P2Y12 inhibitors). Oral beta-blocker early only if no heart failure, hypotension or block.

5

Before discharge (secondary prevention)

Dual antiplatelet therapy for 12 months, high-intensity statin (atorvastatin 40–80 mg), beta-blocker, angiotensin-converting enzyme (ACE) inhibitor (or angiotensin receptor blocker (ARB)), stop smoking, cardiac rehabilitation, control diabetes and blood pressure.

NON-ST-ELEVATION ACUTE CORONARY SYNDROME (NSTE-ACS) (NON-ST-ELEVATION MYOCARDIAL INFARCTION (NSTEMI) AND UNSTABLE ANGINA): TIMING OF ANGIOGRAPHY DEPENDS ON RISK

Immediate (<2 h): hemodynamic instability or shock, ongoing pain despite treatment, life-threatening arrhythmia, acute heart failure, mechanical complication, recurrent dynamic ST changes.
Early (<24 h): confirmed NSTEMI, dynamic ST/T changes, transient ST elevation, Global Registry of Acute Coronary Events (GRACE) score >140.
Selective: low risk → non-invasive testing first.
Treat with aspirin + anticoagulant (enoxaparin, fondaparinux or unfractionated heparin). P2Y12 inhibitor before angiography is not routine if early angiography is planned.

TRAPS

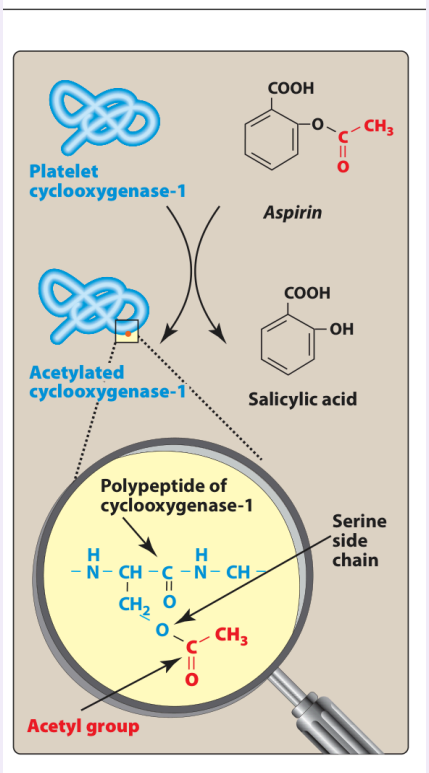
- **Inferior ST-elevation myocardial infarction (STEMI)**: do a right-sided electrocardiogram (ECG) (V4R). Right ventricular infarction is preload-dependent → **no nitrates**, give fluids if hypotensive.
- **Aortic dissection can cause an inferior STEMI** (flap blocks the right coronary). Giving fibrinolytics or heparin is fatal → if the pain was tearing or there is a pulse difference, image the aorta first.
- **Cocaine chest pain**: benzodiazepines first; avoid beta-blockers in the acute phase.

The drugs (each box complete on its own)

DRUG BOX

Aspirin (acetylsalicylic acid)

Class	Antiplatelet, cyclooxygenase (COX) inhibitor
Mechanism	**Irreversibly acetylates COX-1** → platelets cannot make thromboxane A2 (a platelet activator and vasoconstrictor). Platelets have no nucleus, so they cannot make new COX → effect lasts the platelet lifespan, **7–10 days**.
Use here	Every acute coronary syndrome (ACS) unless allergic. Loading 150–300 mg chewed (non-enteric-coated), then 75–100 mg daily for life.
Side effects	Gastrointestinal ulcer and bleeding, aspirin-exacerbated asthma, tinnitus at high doses, Reye syndrome in children.
Avoid	Active bleeding, true allergy.

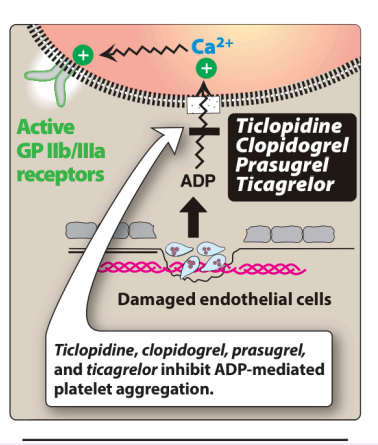


Acetylation of cyclooxygenase-1 by aspirin. *Lippincott Illustrated Reviews: Pharmacology, Fig. 22.7.*

DRUG BOX

P2Y12 inhibitors: clopidogrel, prasugrel, ticagrelor

Mechanism	Block the **adenosine diphosphate (ADP) receptor P2Y12** on platelets → the glycoprotein IIb/IIIa (GP IIb/IIIa) receptor is not activated → no aggregation. Given with aspirin = **dual antiplatelet therapy (DAPT)**.
Clopidogrel	Prodrug activated by liver cytochrome P450 2C19 (CYP2C19) → weaker, variable (poor metabolizers, omeprazole interaction). Irreversible. 300–600 mg load, then 75 mg daily. Choice with fibrinolysis or high bleeding risk.
Prasugrel	Prodrug, stronger, irreversible. 60 mg load, then 10 mg daily (5 mg if ≥75 years or <60 kg). **Contraindicated after stroke or transient ischemic attack (TIA).**
Ticagrelor	Not a prodrug, **reversible**, twice daily. 180 mg load, then 90 mg twice daily. Side effects: breathlessness and bradycardia (raises adenosine).
Side effects	Bleeding. Stop before bypass surgery: clopidogrel and ticagrelor 5 days, prasugrel 7 days.



Mechanism of P2Y12 inhibitors. *Lippincott Pharmacology, Fig. 22.8.*

DRUG BOX

Heparins: unfractionated heparin (UFH), low molecular weight heparin (LMWH, enoxaparin), fondaparinux

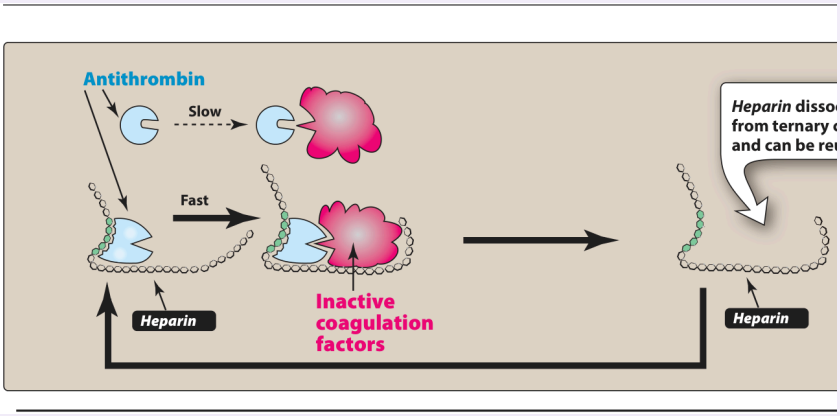
Mechanism Bind **antithrombin** and speed it up about 1000× → antithrombin inactivates **thrombin (IIa) and factor Xa**. UFH chains are long enough to bridge antithrombin to thrombin → inhibits both. LMWH chains are short → **mainly Xa**. Fondaparinux = only the pentasaccharide → **Xa only**.

Use here UFH intravenous (IV) during percutaneous coronary intervention (PCI) (70–100 international units (IU)/kg bolus); monitor activated partial thromboplastin time (aPTT). Enoxaparin 1 mg/kg subcutaneously twice daily or fondaparinux 2.5 mg daily in non-ST-elevation acute coronary syndrome (NSTE-ACS).

Side effects Bleeding; **heparin-induced thrombocytopenia** (antibodies against heparin–platelet factor 4, day 5–10, platelets fall >50% and paradoxical clots); osteoporosis with long use.

Kidneys LMWH and fondaparinux are renally cleared: reduce dose or use UFH if creatinine clearance <30 mL/min.

Reversal **Protamine sulfate**: full for UFH, partial for LMWH, none for fondaparinux.



Heparin accelerates the inactivation of clotting factors by antithrombin. *Lippincott Pharmacology, Fig. 22.12.*

DRUG BOX

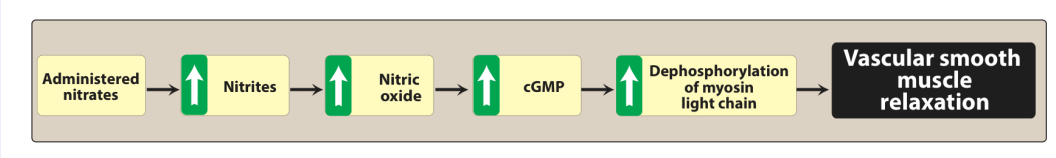
Nitrates: glyceryl trinitrate (nitroglycerin), isosorbide mononitrate / dinitrate

Mechanism Release **nitric oxide** → activates guanylyl cyclase → ↑ cyclic guanosine monophosphate (cGMP) → dephosphorylation of myosin light chain → smooth muscle relaxation. Mainly **veins** → less venous return (↓ preload) → smaller, less stretched ventricle → **↓ oxygen demand**. Also dilates coronary arteries (relieves spasm).

Use here Sublingual 0.4 mg (or 400 µg spray) every 5 min up to 3 doses, or intravenous (IV) infusion for ongoing pain, hypertension or heart failure. Long-acting forms for stable angina.

Side effects Headache, flushing, hypotension, reflex tachycardia, tolerance (needs a nitrate-free interval).

Contraindicated Systolic blood pressure (BP) <90 mmHg, **right ventricular infarction**, phosphodiesterase-5 (PDE-5) inhibitor in last 24 h (sildenafil) or 48 h (tadalafil), severe aortic stenosis, hypertrophic obstructive cardiomyopathy.



Effect of nitrates on smooth muscle. *Lippincott Pharmacology, Fig. 21.5.*

DRUG BOX

Beta-blockers: metoprolol, bisoprolol, atenolol

Mechanism Block β1 receptors in the heart → ↓ heart rate and ↓ contractility → **↓ oxygen demand**. Slower rate = longer diastole = **more time for coronary filling**. Antiarrhythmic (fewer ventricular arrhythmias). Reduce mortality after myocardial infarction (MI).

Use here Oral, within 24 h, if stable (e.g., metoprolol 25–50 mg). Long-term after MI, especially if ejection fraction ≤40%. intravenous (IV) only for ongoing ischemia with hypertension or tachycardia.

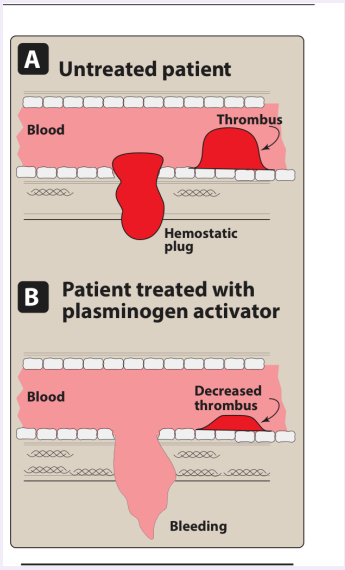
Contraindicated Acute heart failure, cardiogenic shock, bradycardia or heart block, severe asthma, vasospastic angina, acute cocaine toxicity.

Side effects Bradycardia, hypotension, fatigue, bronchospasm, masks hypoglycemia.

DRUG BOX

Fibrinolytics: tenecteplase, alteplase, reteplase (streptokinase in some countries)

Mechanism	Convert **plasminogen → plasmin**, which digests the **fibrin** of the clot. Tissue plasminogen activators (alteplase, tenecteplase) are fibrin-selective. Streptokinase is not fibrin-selective and is antigenic (allergy, cannot repeat).
Use here	ST-elevation myocardial infarction (STEMI) when percutaneous coronary intervention (PCI) cannot be done within 120 min. Tenecteplase single weight-based intravenous (IV) bolus (half dose if ≥75 years). Also massive pulmonary embolism (PE) with shock (alteplase).
Absolute contraindications	Any previous brain hemorrhage or stroke of unknown cause · ischemic stroke in last 6 months · brain tumor or vascular malformation · major trauma, surgery or head injury in last month · gastrointestinal bleeding in last month · bleeding disorder · **aortic dissection** · non-compressible puncture in last 24 h.
Relative contraindications	transient ischemic attack (TIA) in last 6 months, oral anticoagulant, pregnancy or 1 week postpartum, blood pressure (BP) >180/110, advanced liver disease, endocarditis, active peptic ulcer, long or traumatic cardiopulmonary resuscitation (CPR).



The plasminogen activator dissolves both the unwanted thrombus and the protective hemostatic plug → bleeding. *Lippincott Pharmacology, Fig. 22.20.*

DRUG BOX

Statins: atorvastatin, rosuvastatin

Mechanism Inhibit **3-hydroxy-3-methylglutaryl coenzyme A (HMG-CoA) reductase** (rate-limiting enzyme of cholesterol synthesis) → liver cell cholesterol falls → liver makes more low-density lipoprotein (LDL) receptors → takes LDL out of the blood. Also **stabilize plaques** (less inflammation, thicker cap).

Use here High intensity as early as possible in every acute coronary syndrome (ACS): atorvastatin 40–80 mg or rosuvastatin 20–40 mg. Target LDL <1.4 mmol/L (<55 mg/dL) and ≥50% reduction.

Side effects Muscle pain, rarely myositis and rhabdomyolysis, raised liver enzymes, slight rise in diabetes risk.

DRUG BOX

Morphine and other intravenous (IV) opioids

Mechanism μ-opioid receptor agonist → analgesia, less anxiety → less sympathetic drive and oxygen demand. Mild venodilation.

Use here Small IV titrated doses for severe pain not relieved by nitrates.

Caution Respiratory depression, hypotension, vomiting; **slows gut absorption of oral P2Y12 inhibitors**.

DRUG BOX

angiotensin-converting enzyme (ACE) inhibitors (ramipril, lisinopril) / angiotensin receptor blockers (ARB)

Mechanism Block angiotensin II production (or its receptor) → less vasoconstriction, less aldosterone → lower afterload and less **remodeling** (dilation and scarring) of the infarcted ventricle.

Use here After myocardial infarction (MI), especially with heart failure, ejection fraction ≤40%, diabetes, hypertension or chronic kidney disease.

Side effects Cough and angioedema (ACE inhibitors, from bradykinin), hyperkalemia, rise in creatinine, hypotension. Contraindicated in pregnancy.

7.2 Aortic Dissection KILLER

Aortic dissection: blood tears through the inner layer of the aorta (intima) and splits the wall (media), creating a **false lumen**. Untreated type A dissection has a mortality of about 1–2% per hour in the first day.

RETROGRADE CHAIN

SYMPTOM **Sudden, worst-ever, tearing pain in the chest, moving to the back between the shoulder blades and down**

▼ WHY? ▼

PHYSIOLOGY The aortic wall is being stretched and torn as the dissection propagates (the adventitia has pain fibres) → pain is maximal at the moment of tearing and **migrates** with the tear

▼ WHY? ▼

ANATOMY The false lumen runs along the aorta and can **block the origin of any branch**: subclavian (pulse and blood pressure (BP) difference), carotid (stroke), coronary (myocardial infarction (MI), usually right coronary artery (RCA) → inferior), spinal (paraplegia), mesenteric (gut ischemia), renal (kidney failure). Backwards into the root: **aortic regurgitation** and **tamponade** (main cause of death)

▼ WHY? ▼

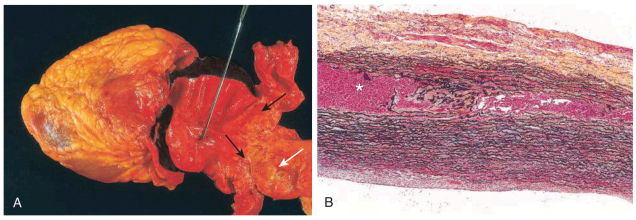
EVENT An **intimal tear** lets blood under pressure enter the media and dissect between its layers. Most tears occur where the wall stress is highest: ascending aorta just above the valve, or just after the left subclavian artery

▼ WHY? ▼

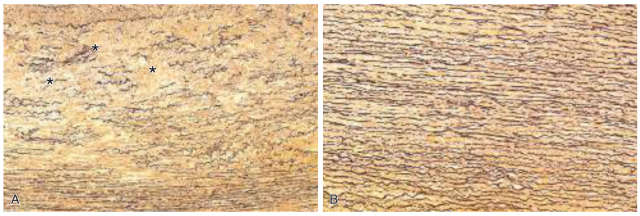
MICROSCOPY **Cystic medial degeneration**: loss and fragmentation of elastic fibres, smooth muscle loss, pools of mucoid ground substance → weak media

▼ WHY? ▼

CAUSE **Hypertension** (most common: chronic stress injures the vasa vasorum and the media) · **Marfan syndrome** (fibrillin-1 mutation → abnormal elastic fibres and excess transforming growth factor beta (TGF-β) signalling) · Ehlers-Danlos type intravenous (IV) · bicuspid aortic valve · Turner syndrome · cocaine (sudden BP surge) · pregnancy · aortic instrumentation



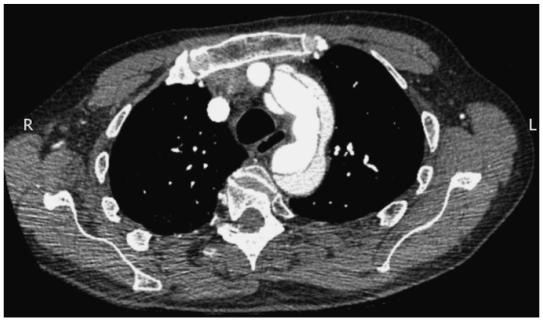
Aortic dissection. (A) Opened aorta: the probe passes through the intimal tear into the false lumen filled with hematoma. (B) Histology: hematoma (asterisk) splitting the media, elastic layers black. *Robbins, Fig. 10.20.*



Cystic medial degeneration (A, Marfan) vs normal media (B). Elastic fibres stained black are fragmented with empty "cystic" spaces. *Robbins, Fig. 10.18.*

Classification decides the treatment

Stanford	Involves	Treatment
Type A	Ascending aorta (± beyond)	**Emergency surgery**
Type B	Descending aorta only (distal to left subclavian)	Medical (blood pressure (BP) and heart rate control); endovascular stent (thoracic endovascular aortic repair (TEVAR)) if complicated: rupture, malperfusion, uncontrolled pain or BP



computed tomography (CT) showing dissection of the aortic arch: the intimal flap separates true and false lumen. *Oxford Handbook of Emergency Medicine (OHEM), Fig. 3.23.*

Diagnosis

- Aortic Dissection Detection Risk Score (ADD-RS) (section 5): 0–1 with negative D-dimer → unlikely; ≥2 → image directly.
- **computed tomography (CT) aortography**: test of choice if stable. Transesophageal echo if unstable.
- chest X-ray (CXR): widened mediastinum in about 60%, often normal. electrocardiogram (ECG): normal or ischemic.

Management

1. **Pain relief with intravenous (IV) opioid**
   (pain drives the blood pressure (BP) up).

2. **intravenous (IV) beta-blocker FIRST**
   (esmolol or labetalol): target heart rate **<60** and systolic blood pressure (BP) **100–120 mmHg**. Reduces the force and speed of each ejection (dP/dt), which drives the tear.

3 **Then a vasodilator if blood pressure (BP) still high**
(nitroprusside, nicardipine). Never before the beta-blocker: vasodilation alone causes reflex tachycardia and more shear stress.

4 **Call cardiothoracic / vascular surgery**
: type A to theatre; type B to intensive care.

NEVER

Do not give fibrinolytics, heparin or antiplatelets until dissection is excluded in a patient with tearing pain, pulse deficit or a wide mediastinum, even if the electrocardiogram (ECG) shows ST elevation.

DRUG BOX

Esmolol (intravenous (IV)) and labetalol (IV)

Mechanism	Esmolol: ultra-short-acting selective β1 blocker (half-life ~9 min, broken down by red-cell esterases) → easy to titrate and stop. Labetalol: α1 + non-selective β blocker → lowers rate and blood pressure (BP) together.
Use here	First-line rate and BP control in aortic dissection; also hypertensive emergencies.
Avoid	Bradycardia, heart block, decompensated heart failure, severe asthma (labetalol). If beta-blockers are contraindicated: non-dihydropyridine calcium channel blocker (diltiazem, verapamil).

7.3 Pulmonary Embolism (PE) KILLER

Pulmonary embolism: a thrombus, almost always from the deep veins of the legs or pelvis, travels through the right heart and blocks a pulmonary artery.

RETROGRADE CHAIN: THREE CONSEQUENCES, THREE GROUPS OF SYMPTOMS

SYMPTOMS **Sudden breathlessness, pleuritic chest pain, tachycardia; sometimes hemoptysis, fainting or shock**

▼ WHY? ▼

CONSEQUENCE 1 **Dead space**: part of the lung is ventilated but not perfused → ventilation/perfusion mismatch → **low oxygen** → fast breathing → CO_2 washed out → respiratory alkalosis

▼ WHY? ▼

CONSEQUENCE 2 **Right heart overload**: pulmonary vascular resistance rises → the thin right ventricle dilates and fails → septum bows into the left ventricle → left ventricle underfilled → ↓ cardiac output → **hypotension, syncope**; raised jugular venous pressure (JVP); right ventricular ischemia → raised troponin

▼ WHY? ▼

CONSEQUENCE 3 **Pulmonary infarction** (only ~10%, because the bronchial arteries also feed the lung): small peripheral wedge-shaped hemorrhagic infarct touching the pleura → **pleuritic pain, hemoptysis**, small effusion

▼ WHY? ▼

EVENT Deep vein thrombus breaks free → passes vena cava, right atrium, right ventricle → lodges in the pulmonary arteries (large clot can sit at the bifurcation: **saddle embolus**)

▼ WHY? ▼

CAUSE **Virchow triad: stasis (immobility, surgery, long travel, heart failure) · hypercoagulability (cancer, estrogen, pregnancy, inherited thrombophilia such as factor V Leiden) · endothelial injury (surgery, trauma, previous clot)**

Diagnostic pathway

1. **Unstable (shock)?**
   → bedside echo: dilated right ventricle → treat as high-risk pulmonary embolism (PE) (computed tomography (CT) if possible).

2. **Stable: estimate probability**
   with Wells (section 5). If low probability → Pulmonary Embolism Rule-out Criteria (PERC).

3. **pulmonary embolism (PE) unlikely**
   → D-dimer (age-adjusted). Negative = PE excluded. Positive → computed tomography (CT) pulmonary angiography.

4. **pulmonary embolism (PE) likely**
   → computed tomography (CT) pulmonary angiography directly. Ventilation/perfusion scan if contrast is contraindicated or in pregnancy.



Embolus from a leg deep vein thrombus lodged in a pulmonary artery branch. *Robbins, Fig. 4.15.*

Management by risk

Risk	Definition	Treatment
High (massive)	Shock or systolic blood pressure (BP) <90 mmHg	unfractionated heparin (UFH) + **systemic thrombolysis** (alteplase 100 mg over 2 h); surgical or catheter clot removal if thrombolysis is contraindicated
Intermediate	Stable, but right ventricular dysfunction and/or raised troponin	Anticoagulation, monitor closely; rescue thrombolysis if deteriorates
Low	Stable, no RV dysfunction (low Pulmonary Embolism Severity Index (PESI) score)	Anticoagulation; early discharge possible

DRUG BOX

Direct oral anticoagulants (DOACs): apixaban, rivaroxaban (factor Xa inhibitors), dabigatran (thrombin inhibitor)

Mechanism	Directly bind and block **factor Xa** (apixaban, rivaroxaban) or **thrombin** (dabigatran). No antithrombin needed, no routine monitoring.
Use here	First choice for pulmonary embolism (PE) in most patients. Apixaban 10 mg twice daily for 7 days, then 5 mg twice daily. Rivaroxaban 15 mg twice daily for 21 days, then 20 mg daily with food. At least 3 months; longer if unprovoked or ongoing risk.
Not for	Pregnancy and breastfeeding (use low molecular weight heparin (LMWH)), mechanical heart valves, antiphospholipid syndrome, severe kidney failure.
Side effects	Bleeding. Reversal: andexanet alfa (Xa inhibitors), idarucizumab (dabigatran), or prothrombin complex concentrate.

DRUG BOX

Heparins: unfractionated heparin (UFH) and low molecular weight heparin (LMWH)

Mechanism	Activate antithrombin → inactivates thrombin and factor Xa (UFH); mainly Xa (LMWH).
Use here	LMWH (enoxaparin 1 mg/kg twice daily) when direct oral anticoagulants (DOACs) are not suitable, in pregnancy, and often in cancer. **UFH** if thrombolysis may be needed, severe kidney failure, or high bleeding risk (short half-life, fully reversible with protamine).
Watch	Bleeding, heparin-induced thrombocytopenia (check platelets).

DRUG BOX

Alteplase (recombinant tissue plasminogen activator) for high-risk pulmonary embolism (PE)

Mechanism	Converts **plasminogen → plasmin** on the surface of fibrin → plasmin digests the fibrin of the clot → the pulmonary artery reopens and the right ventricle is unloaded within hours.
Use here	Only **high-risk PE** (shock or systolic blood pressure (BP) <90 mmHg): 100 mg intravenous (IV) over 2 hours (accelerated 0.6 mg/kg over 15 min in cardiac arrest). Stop unfractionated heparin (UFH) during the infusion, restart after.
Contraindicated	Previous brain hemorrhage, ischemic stroke in last 6 months, brain tumor, major trauma or surgery or head injury in last 3 weeks, active bleeding, bleeding disorder. In a dying patient the absolute ones become relative.
Side effects	Bleeding, including intracranial hemorrhage (about 2%).

7.4 Tension Pneumothorax KILLER

Tension pneumothorax: air enters the pleural space and cannot leave, so the pressure rises with every breath. It is a **clinical diagnosis**; do not wait for an X-ray.

RETROGRADE CHAIN

SYMPTOM **Sudden one-sided pleuritic pain, severe breathlessness, then collapse**

▼ WHY? ▼

SIGNS Absent breath sounds and hyperresonance on one side, trachea deviated **away**, raised jugular venous pressure (JVP), hypotension, tachycardia, hypoxia

▼ WHY? ▼

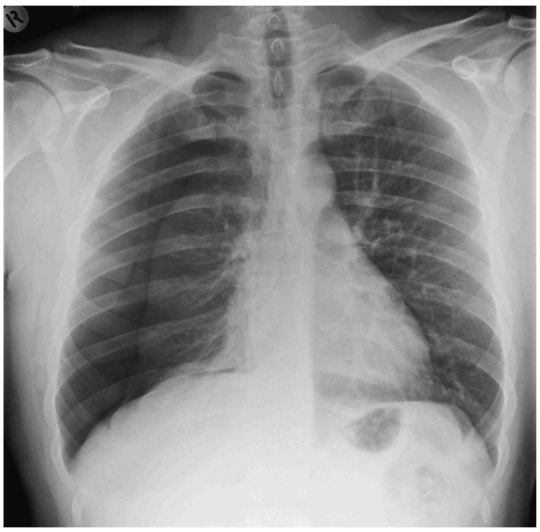
HEMODYNAMICS Mediastinum pushed to the other side → **vena cava kinked and compressed** → venous return falls → cardiac output falls → **obstructive shock**

▼ WHY? ▼

MECHANISM **One-way valve**: air enters on inspiration, cannot leave on expiration → intrapleural pressure becomes positive → lung collapses

▼ WHY? ▼

CAUSE **Trauma · positive-pressure ventilation · primary spontaneous (rupture of apical subpleural blebs: tall thin young men, smokers) · secondary (chronic obstructive pulmonary disease (COPD), cystic fibrosis, pneumonia) · procedures (central line, pleural tap)**



Spontaneous right pneumothorax (simple, not tension): visible lung edge, no lung markings beyond it. In tension the mediastinum is shifted to the left. *Oxford Handbook of Emergency Medicine (OHEM), Fig. 3.28.*

Management

1. **High-flow oxygen and immediate decompression**
   Needle or finger thoracostomy: **4th–5th intercostal space, just anterior to the mid-axillary line** (Advanced Trauma Life Support (ATLS) 10th edition; adults). Second intercostal space in the mid-clavicular line is the alternative (and in children).

2 **Chest drain**
in the "safe triangle" (5th intercostal space, between anterior and mid-axillary lines) straight after.

Simple (non-tension) spontaneous pneumothorax: small and asymptomatic → observe; symptomatic or large → aspiration or drain. Advise no flying until resolved and never scuba diving unless definitive surgery.

7.5 Cardiac Tamponade KILLER

RETROGRADE CHAIN

SYMPTOM **Breathlessness, chest discomfort, dizziness → collapse**

▼ WHY? ▼

SIGNS **Beck triad**: hypotension, raised jugular venous pressure (JVP), muffled heart sounds · **pulsus paradoxus** · tachycardia · electrocardiogram (ECG) low voltage and electrical alternans

▼ WHY? ▼

HEMODYNAMICS Pressure in the pericardium exceeds the filling pressure of the right heart → right atrium and ventricle **collapse in diastole** → cannot fill → stroke volume falls. The heart can only fill at the expense of the other side: inspiration fills the right → septum bulges left → **left output drops on inspiration** (pulsus paradoxus)

▼ WHY? ▼

KEY FACT The pericardium is stiff. **Speed matters more than volume**: 150 mL quickly (trauma, rupture) can kill; 1 L slowly (cancer, uremia) may be tolerated

▼ WHY? ▼

CAUSE **Trauma · free wall rupture after myocardial infarction (MI) · type A aortic dissection · cancer · uremia · tuberculosis · viral or autoimmune pericarditis · after cardiac surgery or procedure · anticoagulation**

MANAGEMENT

Urgent echo → **pericardiocentesis** (ultrasound-guided, subxiphoid), or surgical drainage for trauma, rupture or dissection. intravenous (IV) fluids to support filling while waiting. **Avoid** diuretics, nitrates and positive-pressure ventilation if possible (they reduce filling further).

7.6 Esophageal Rupture (Boerhaave syndrome) and Perforated Peptic Ulcer

KILLER

RETROGRADE CHAIN: ESOPHAGEAL RUPTURE

SYMPTOM **Severe retrosternal pain after forceful vomiting, then fever, breathlessness and shock**

▼ WHY? ▼

SIGNS **Mackler triad**: vomiting, chest pain, subcutaneous emphysema · Hamman crunch · left pleural effusion

▼ WHY? ▼

PATHOLOGY Gastric acid, food and bacteria spill into the mediastinum → chemical then bacterial **mediastinitis** → sepsis (mortality rises sharply after 24 h)

▼ WHY? ▼

EVENT Vomiting against a closed upper esophageal sphincter → sudden rise in pressure → **full-thickness tear**, usually the left posterolateral wall of the distal esophagus (the weakest point)

▼ WHY? ▼

CAUSE **Binge drinking or eating with retching · iatrogenic (endoscopy, the most common cause overall) · trauma · foreign body**

DIAGNOSIS AND TREATMENT: RUPTURE

chest X-ray (CXR): pneumomediastinum, left effusion. **computed tomography (CT) with water-soluble oral contrast** or contrast swallow.
Nil by mouth, intravenous (IV) broad-spectrum antibiotics + antifungal, IV proton pump inhibitor, drainage, early surgery or endoscopic stent.

PERFORATED PEPTIC ULCER

Sudden epigastric pain spreading over the abdomen, rigid board-like abdomen, history of ulcer or non-steroidal anti-inflammatory drug (NSAID) use. **Erect chest X-ray (CXR): free air under the diaphragm**; computed tomography (CT).
Treatment: resuscitation, nil by mouth, intravenous (IV) proton pump inhibitor (PPI), antibiotics, surgical repair.

DRUG BOX

Proton pump inhibitors (PPI): omeprazole, pantoprazole, esomeprazole

Mechanism	Prodrugs activated in the acid of the parietal cell → **irreversibly block the H^+/K^+-ATPase** (the final step of acid secretion). Take 30 minutes before a meal, when the pumps are active.
Use here	intravenous (IV) in perforation and bleeding; oral for gastroesophageal reflux disease (GERD) (omeprazole 20 mg daily for 4–8 weeks) and for stomach protection with antiplatelets.
Side effects	Long-term: low magnesium, B12 deficiency, C. difficile infection, fractures. Omeprazole reduces clopidogrel activation (prefer pantoprazole).

7.7 Pericarditis and Myocarditis USUALLY BENIGN

RETROGRADE CHAIN: ACUTE PERICARDITIS

SYMPTOM **Sharp central pain, worse breathing in and lying flat, better sitting forward, radiating to the trapezius ridge**

▼ WHY? ▼

SIGNS **Friction rub** · fever · electrocardiogram (ECG): widespread concave ST elevation + PR depression

▼ WHY? ▼

PHYSIOLOGY The inflamed parietal pericardium has somatic pain fibres and the phrenic nerve (C3–C5) → sharp pain, shoulder-tip referral. Sitting forward moves the heart away from the inflamed back pericardium. Inflammation of the outer (epicardial) layer of the myocardium causes the diffuse ST change

▼ WHY? ▼

PATHOLOGY Acute inflammation with **fibrinous exudate** ("bread and butter" appearance) ± effusion

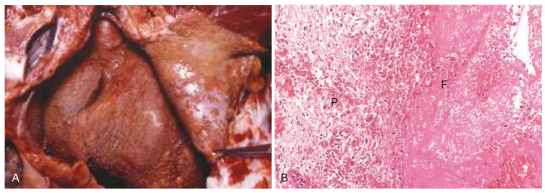
▼ WHY? ▼

CAUSE **Viral** (coxsackie B, echovirus) or idiopathic, the commonest · **tuberculosis** (common where TB is endemic) · after myocardial infarction (MI) (early, day 1–3; Dressler syndrome at weeks) · after cardiac surgery · uremia · autoimmune (lupus, rheumatoid arthritis) · cancer · radiation

DIAGNOSIS: AT LEAST 2 OF 4

- Typical pericarditic chest pain
- Pericardial friction rub
- New widespread ST elevation or PR depression
- New or worsening pericardial effusion

Supporting: raised C-reactive protein (CRP), effusion on echo. Raised troponin = **myopericarditis**.



Fibrinous pericarditis. (A) Fibrin deposits on the pericardium. (B) Pink fibrin mesh (F) over the pericardial surface (P). This is the source of the friction rub. *Robbins, Fig. 3.13.*

ADMIT IF

Fever >38 °C, subacute onset, large effusion or tamponade, no response to non-steroidal anti-inflammatory drug (NSAID) after 1 week, immunosuppression, anticoagulation, trauma, myocarditis.

DRUG BOX

non-steroidal anti-inflammatory drugs (NSAIDs) / high-dose aspirin + colchicine (first-line for pericarditis)

NSAID	Ibuprofen 600 mg every 8 h, or aspirin 750–1000 mg every 8 h (preferred **after myocardial infarction (MI)**, because other NSAIDs impair scar healing), for 1–2 weeks then taper. Inhibit cyclooxygenase (COX) → less prostaglandin → less inflammation and pain. Give with a proton pump inhibitor (PPI).
Colchicine	0.5 mg twice daily (once daily if <70 kg) for **3 months**: halves the recurrence rate. Binds tubulin → blocks microtubules → neutrophils cannot migrate; inhibits the NOD-like receptor protein 3 (NLRP3) inflammasome.
Side effects	NSAID: stomach ulcer, kidney injury. Colchicine: diarrhea, toxicity with cytochrome P450 3A4 (CYP3A4) / P-glycoprotein inhibitors (clarithromycin) and kidney failure.
Steroids	Second line only (raise recurrence), unless autoimmune cause or contraindication to NSAIDs. Restrict exercise until symptoms and C-reactive protein (CRP) resolve (athletes ≥3 months).

MYOCARDITIS IN ONE PARAGRAPH

Inflammation of the myocardium itself, mostly viral (also drugs, autoimmune, immune checkpoint inhibitors, after messenger RNA (mRNA) vaccine in young men). Chest pain + **raised troponin** + electrocardiogram (ECG) changes in a young patient after a viral illness, with **normal coronaries**. Can cause heart failure, arrhythmias and sudden death. Diagnosis: echo, **cardiac magnetic resonance imaging (MRI)**, sometimes biopsy. Treatment: supportive, heart failure therapy, no exercise for 3–6 months.

7.8 Stable Angina and Vasospastic Angina CHRONIC

STABLE ANGINA

SYMPTOM **Predictable pressure with exertion, gone within minutes of rest or nitrate**

▼ WHY? ▼

PHYSIOLOGY Exercise raises oxygen demand; a fixed narrowing (>70%) cannot increase flow → temporary **subendocardial ischemia** → ST depression during the pain only

▼ WHY? ▼

CAUSE **Stable plaque**: thick fibrous cap, narrows the lumen slowly, does not rupture

VASOSPASTIC (PRINZMETAL) ANGINA

SYMPTOM **Pain at rest, often early morning, young patient, smoker**

▼ WHY? ▼

PHYSIOLOGY Coronary artery goes into **spasm** → total temporary occlusion → **transient ST elevation** that resolves with the pain

▼ WHY? ▼

CAUSE **Hyperreactive smooth muscle and endothelial dysfunction; triggers: smoking, cocaine, triptans, cold, hyperventilation**

	Stable angina	Vasospastic angina
Diagnosis	Stress testing (exercise electrocardiogram (ECG), stress echo, perfusion scan) or computed tomography (CT) coronary angiography; angiography if high risk	Clinical + ECG during pain; normal or near-normal coronaries; provocation test
Symptom relief	Sublingual nitrate + **beta-blocker** or calcium channel blocker; then long-acting nitrate, ranolazine, ivabradine	**Calcium channel blocker** + nitrates; stop smoking
Prevent events	Aspirin, statin, angiotensin-converting enzyme (ACE) inhibitor if indicated, risk factor control	Statin and risk factor control
Avoid		**Non-selective beta-blockers (unopposed α vasoconstriction worsens spasm)**

DRUG BOX

Calcium channel blockers: amlodipine (dihydropyridine), diltiazem and verapamil (non-dihydropyridines)

Mechanism	Block L-type calcium channels. Dihydropyridines (amlodipine) act mainly on **arterial smooth muscle** → vasodilation, relieve spasm. Diltiazem and verapamil also act on the **heart** → slower rate, less contractility → ↓ oxygen demand.
Use here	First line in vasospastic angina; alternative or add-on to beta-blockers in stable angina.
Side effects	Amlodipine: ankle edema, headache, flushing. Verapamil/diltiazem: bradycardia, heart block, constipation; do not combine with beta-blockers or use in heart failure with reduced ejection fraction.

7.9 The benign causes AFTER KILLERS EXCLUDED

Cause	Mechanism → symptom	Management
Gastroesophageal reflux disease (GERD)	Weak lower esophageal sphincter → acid refluxes → irritates esophageal mucosa → burning retrosternal pain, worse lying and after meals	Weight loss, avoid late meals, raise bed head; proton pump inhibitor (PPI) for 4–8 weeks. **Alarm features** (dysphagia, weight loss, bleeding, anemia, age >55 with new symptoms) → endoscopy
Esophageal spasm	Uncoordinated strong contractions of the esophageal muscle → squeezing pain that mimics angina, also relieved by nitrates	Exclude cardiac cause; calcium channel blocker, nitrate, treat reflux
Costochondritis (Tietze if swollen)	Inflammation of costal cartilage joints → focal sharp pain reproduced by pressing	Reassurance, non-steroidal anti-inflammatory drugs (NSAIDs), heat
Herpes zoster	Varicella reactivates in one dorsal root ganglion → burning band of pain in one dermatome → vesicles 2–3 days later	Valaciclovir 1 g three times daily for 7 days, within 72 h of rash; analgesia
Pneumonia / pleurisy	Infection reaches the parietal pleura → pleuritic pain, fever, cough	Antibiotics by severity (Confusion, Urea, Respiratory rate, Blood pressure, age ≥65 (CURB-65))
Panic attack	Sympathetic surge and hyperventilation → chest tightness, palpitations, tingling (low CO_2 lowers ionized calcium)	Only after excluding killers; reassurance, breathing techniques, cognitive behavioural therapy (CBT), selective serotonin reuptake inhibitor (SSRI) if recurrent

BEFORE YOU WRITE "NON-CARDIAC CHEST PAIN"

Ask yourself: have I actively excluded each of the six killers, with the right test for each? Is the troponin **serial**? Is there a safe follow-up plan and clear return advice?

Credits and sources

Source	Used for
Wyatt JP, Taylor RG, et al. *Oxford Handbook of Emergency Medicine*, 5th ed. Oxford University Press.	Emergency approach, differential table, History, electrocardiogram (ECG), Age, Risk factors, Troponin (HEART) score, ECG figures (3.5–3.10), computed tomography (CT) dissection (3.23), chest X-ray (CXR) pneumothorax (3.28)
First Aid Clinical Algorithms for the USMLE Step 2 CK, 2024. McGraw Hill.	Structure of the map (ischemic / non-ischemic cardiac / non-cardiac), ACS algorithm, post-MI complications, coronary territories
Kumar V, Abbas AK, Aster JC. *Robbins Basic Pathology*, 10th ed. Elsevier.	Atherosclerosis, plaque, MI microscopy and timing, dissection, cystic medial degeneration, PE, fibrinous pericarditis; Figures 3.13, 4.15, 10.14–10.20, 11.8–11.13
Whalen K, et al. *Lippincott Illustrated Reviews: Pharmacology*. Wolters Kluwer.	Drug mechanisms; Figures 21.5, 22.7, 22.8, 22.12, 22.20
Innes JA, et al. *Macleod's Clinical Examination*, 15th ed. Elsevier.	History and examination of chest pain
Oxford Handbook of Clinical Medicine, 10th ed. Oxford University Press.	Cross-checking of clinical features and management
UpToDate: *Approach to the adult with nontraumatic chest pain in the emergency department* (accessed Nov 2026).	Emergency red flags, the list of killers, role of D-dimer, Aortic Dissection Detection Risk Score (ADD-RS) and bedside ultrasound
Byrne RA, et al. 2023 European Society of Cardiology (ESC) Guidelines for the management of acute coronary syndromes. *Eur Heart J* 2023.	ST-elevation myocardial infarction (STEMI) and non-ST-elevation acute coronary syndrome (NSTE-ACS) timing, antithrombotic choice, oxygen threshold
Thygesen K, et al. Fourth Universal Definition of Myocardial Infarction. 2018.	STEMI ECG criteria, definition of MI
Swap CJ, Nagurney JT. Value and limitations of chest pain history in the evaluation of patients with suspected ACS. *JAMA* 2005.	Likelihood ratios of history features
ESC 2019 Pulmonary Embolism and ESC Pericardial Disease guidelines; Advanced Trauma Life Support (ATLS) 10th edition.	PE risk classes and treatment; pericarditis diagnosis and treatment; site of needle decompression

Compiled for personal study by Abdalwadood Al-Khuffash. Written with Claude (Anthropic), November 2026. Doses are typical adult doses; always check local protocols and the patient's kidney function, weight and bleeding risk.